Lab-3 Report

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Sec:5

Questi

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Question 1

(a)

$T_H = \ln(2)\, C (R_1 + R_2) = \ln(2) \times 53 \times 10^{-6} \times (8.2 + 68) = 0.053 \text{ sec}$

$T_2 = \ln(2)\, R_2 C = \ln(2)\,(68) \times 10^{-3} = 0.047 \text{ sec}$

$T_{period} = T_H + T_L = 0.1 \text{ sec}$

$F = 10 \text{ Hz}$

Duty cycle : $D = \frac{T_H}{T_{period}} \times 100 = \frac{0.053}{0.1} \times 100 = 53\%$

(b)

$F = 1 \text{ KHz} \qquad T_{per} = 10^{-3} \text{ sec}$

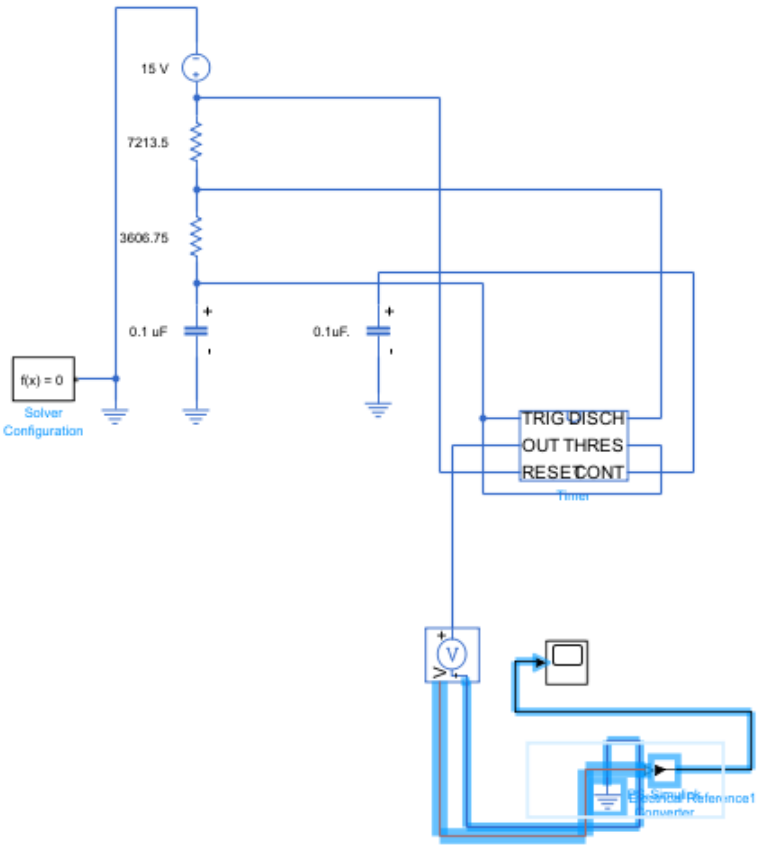
$10^{-3} = T_H + T_L = \ln(2) * 0.1 \times 10^{-6} (R_1 + 2R_2)$

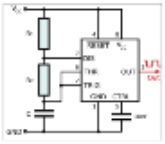
$R_1 + 2R_2 = 14426.95$ ✓ $\qquad D = \frac{T_H}{T_{per}} \times 100 = 75\%$

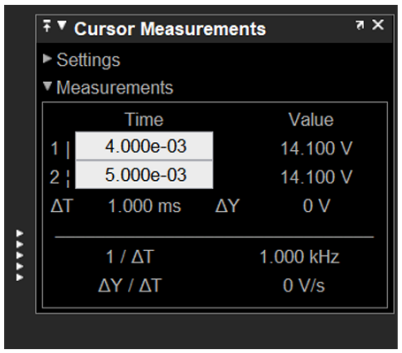
$\frac{\ln 2 \times 0.1 \times 10^{-6} [R_2 + R_1]}{10^{-3}} = 0.75$

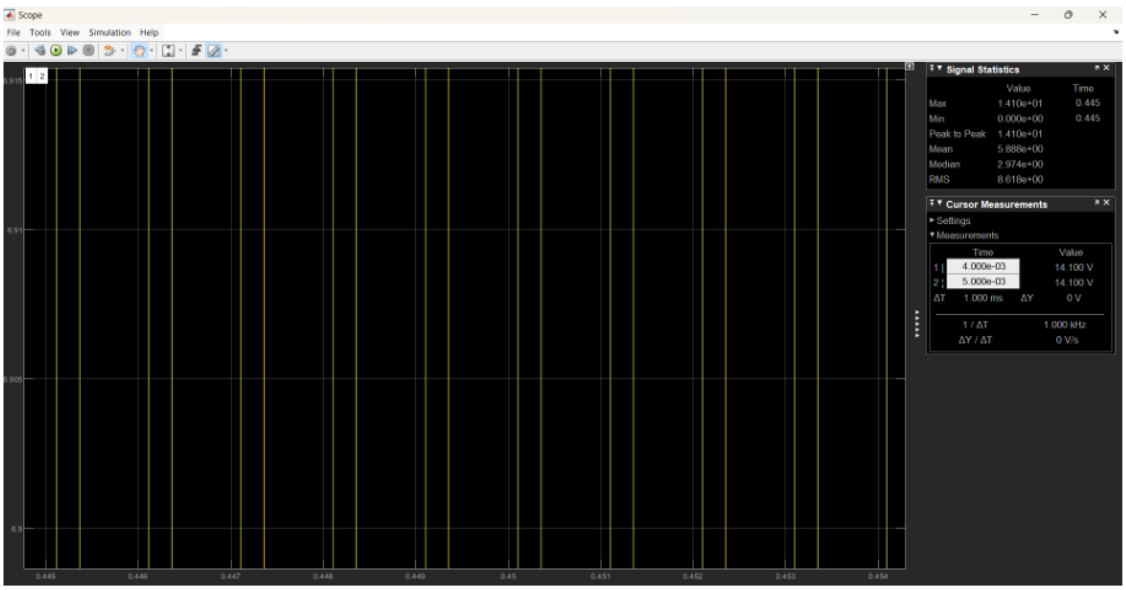
$R_1 + R_2 = 10820.2$

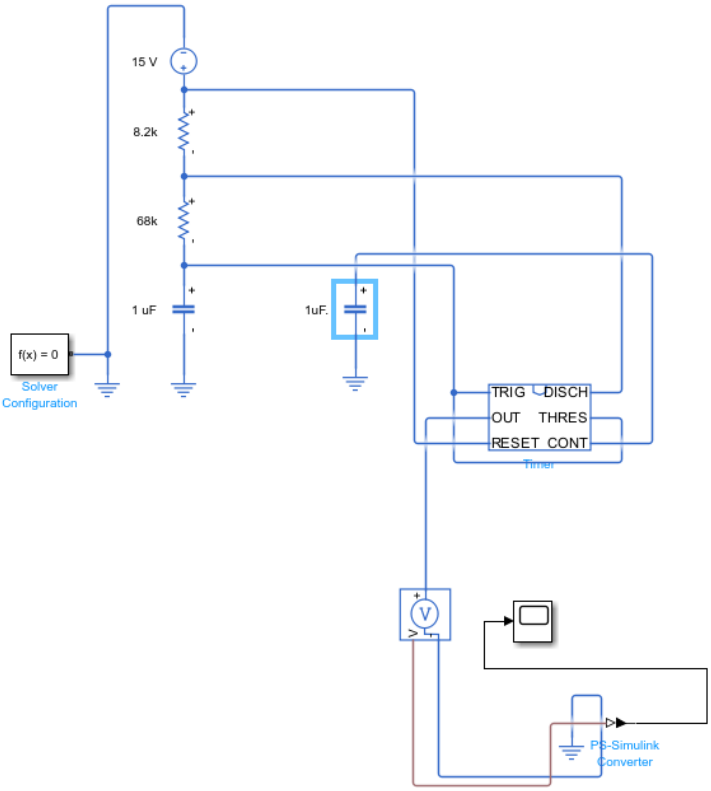
$R_1 = 7213.5$ Ω
$R_2 = 3606.75$ Ω

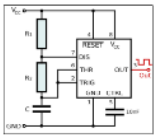

15 V
7213.5
3606.75
0.1 uF
0.1uF.
f(x) = 0
Solver
Configuration
TRIG DISCH
OUT THRES
RESET CONT
Timer

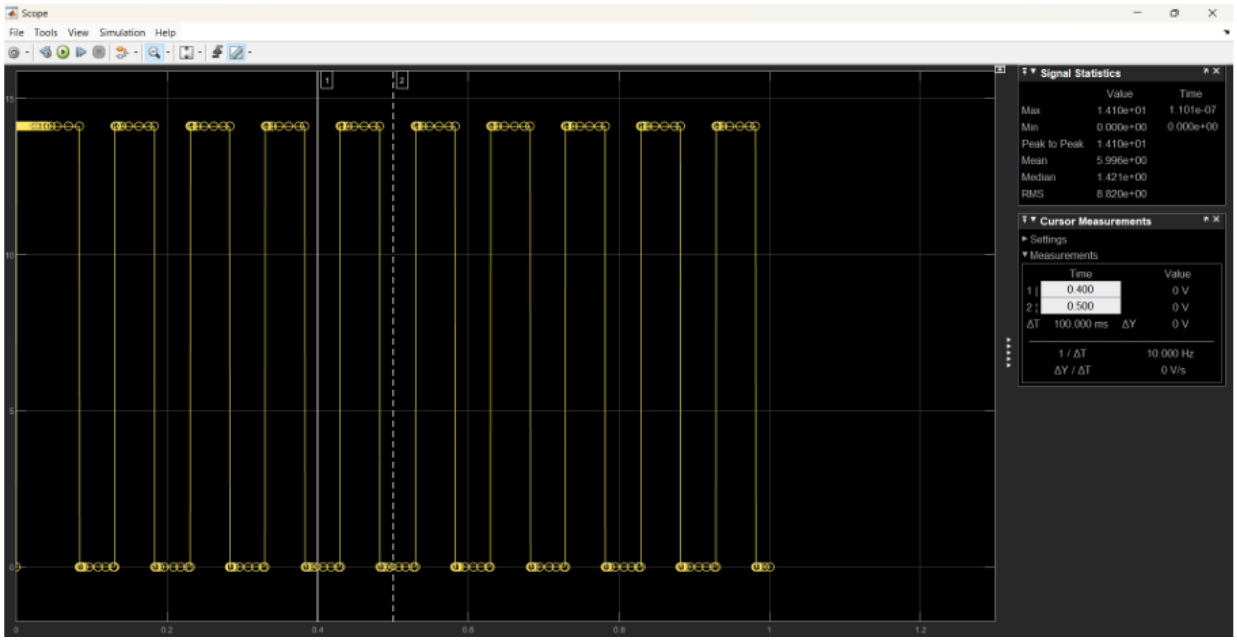



Cursor Measurements
Settings
Measurements
Time
Value
1 | 4.000e-03 14.100 V
2 | 5.000e-03 14.100 V
ΔT 1.000 ms ΔY 0 V
1 / ΔT 1.000 kHz
ΔY / ΔT 0 V/s

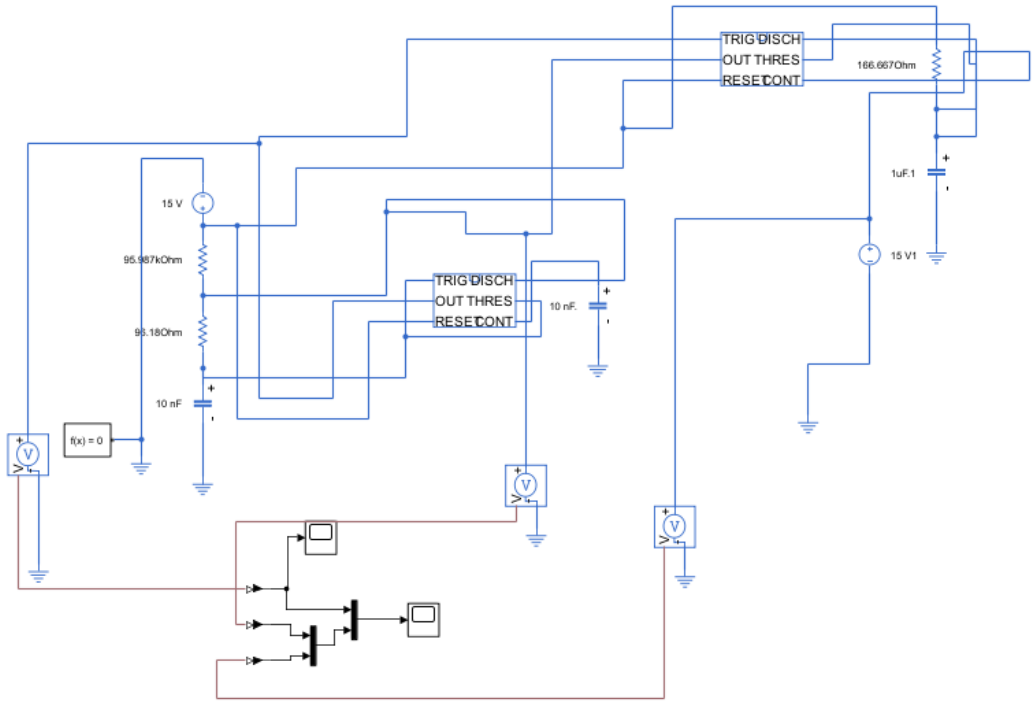

Scope
File Tools View Simulation Help
Signal Statistics
Value Time
Max 1.410e+01 0.445
Min 0.000e+00 0.445
Peak to Peak 1.410e+01
Mean 5.888e+00
Median 2.974e+00
RMS 8.618e+00
Cursor Measurements
Settings
Measurements
Time Value
1 | 4.000e-03 14.100 V
2 | 5.000e-03 14.100 V
ΔT 1.000 ms ΔY 0 V
1 / ΔT 1.000 kHz
ΔY / ΔT 0 V/s


15 V
8.2k
68k
1 uF
1uF.
f(x) = 0
Solver
Configuration
TRIG DISCH
OUT THRES
RESET CONT
Timer
PS-Simulink
Converter




Scope
File Tools View Simulation Help
Signal Statistics
Value Time
Max 1.410e+01 1.101e-07
Min 0.000e+00 0.000e+00
Peak to Peak 1.410e+01
Mean 5.996e+00
Median 1.421e+00
RMS 8.820e+00
Cursor Measurements
Settings
Measurements
Time Value
1 0.400 0 V
2 0.500 0 V
ΔT 100.000 ms ΔY 0 V
1 / ΔT 10.000 Hz
ΔY / ΔT 0 V/s

Question 2


TRIG DISCH
OUT THRES
RESET CONT
166.667Ohm
1uF.1
15 V
95.987kOhm
93.18Ohm
10 nF
f(x) = 0
TRIG DISCH
OUT THRES
RESET CONT
10 nF.
15 V1